MIL-H-22199(Wep)
1 June 1960

MILITARY SPECIFICATION

HELMET, SAILOR'S, REINFORCED PLASTIC, MK 3 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers Helmets, Sailor's Reinforced plastic (Doron), Mk 3 Mod 0, with suspensions, chin straps, and sound attenuating liners, to be used in conjunction with Type SA head-chest telephone set.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

L-P-406	Plastic, Organic: General Specifications, Test Methods
KK-L-311	Leather; Methods of Sampling and Testing
QQ-A-268	Aluminum Alloy 24S Bar, Rod and Wire-Rolled or drawn
QQ-A-355	Aluminum Alloy Plate and Sheet 2024
QQ-A-591	Aluminum Alloy Die Casting
QQ-S-763	Steel Bars, Shapes and Forgings - Corrosion Resisting
QQ-S-766	Steel, Plates, Sheets, Strips - Corrosion Resisting
QQ-W-423	Wire, Steel, Corrosion Resisting
TT-I-599	Ink Marking Stencil, Opaque for Porous Surfaces (Wood Boxes, Fiber Cartons, etc.)
UU-T-116	Tape, Paper, Gummed, Water Resistant
DDD-S-751	Stitches; Seams, and Stitching
PPP-B-601	Boxes, Wood, cleated plywood
PPP-B-621	Boxes, Wood, nailed and lock corner
PPP-B-636	Boxes, Fiber

FSC 8415

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MIL-A-101	Adhesive, Water-resistant, for Sealing Fiberboard Boxes
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in color
MIL-R-7575	Resin, Low Pressure Laminating
MIL-T-7807	Thread, Nylon
MIL-A-8625	Anodic Coating
MIL-F-9084	Fabrics, Woven, Glass, Finished, for Plastic Laminates
MIL-F-10884	Fasteners, Snap
MIL-P-15280	Plastic Foam, Unicellular, Buoyant, Sheet and Molded Shapes
MIL-S-17509	Steel, Casting, Austenitic, Chromium-Nickel, Low Magnetic Permeability
MIL-N-18352	Nylon Plastic, Flexible, Molded or Extruded

STANDARDS

FEDERAL

FED STD 595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

BUWEPS LISTS OF DRAWINGS

LD NO. 535676	Helmet, Sailor's Reinforced Plastic Mk 3 Mod 0

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction samples - Unless otherwise specified in the invitation for bids or contract, preproduction samples of the ballistic shell and complete helmet shall be manufactured using the methods, tools

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and procedures proposed for the production lots. These samples will be tested as specified in Section 4 of this specification, and are for the purpose of determining that the production item meets the design and other requirements of this specification. When samples are approved they will be retained by the contractor for inspection comparison purposes throughout performance of the contract. The size of the preproduction sample shall be as specified in 4.5.

3.2 Materials

3.2.1 The component materials shall be of such quality that will insure the production of helmets which will satisfy the requirements of this specification, and shall be of the same quality as those used in the manufacture of the helmets subjected to the preproduction tests.

3.2.2 Glass cloth - Glass cloth shall conform to Specification MIL-F-9084, Type V.

3.2.3 Bonding resin - The bonding resin shall be rigid polyester resin conforming to specification MIL-R-7575, Type I and shall be approved by the contracting officer for ballistic properties.

3.2.4 Filler - The filler shall be commercial high speed glazing putty.

3.2.5 Sound absorbing material - Foam sound absorbing material shall conform to Specification MIL-P-15280, Type I.

3.2.6 Adhesive - The composition of the adhesive used is not specified herein, but shall meet the requirements specified in 3.5.9.

3.2.7 Protective coating - The protective coating shall be epoxy enamel paint. The color of the protective coating shall match color #36270 of FED STD 595 and shall be capable of meeting the performance characteristics specified in 3.5.7.

3.2.8 Aluminum

3.2.8.1 Coatings - All aluminum parts shall be coated with anodic coating conforming to Specification MIL-A-8625.

3.2.8.2 Aluminum sheet - Aluminum sheet shall conform to Specification QQ-A-355. It shall be formed in the O temper and shall be hardened by heat treatment to T42 temper after fabrication.

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3.2.8.3 Aluminum castings - Aluminum castings shall conform to Specification QQ-A-591, Class 7.

3.2.9 Steel

3.2.9.1 Coating - All steel parts, except the cable, shall be coated with a black oxide coating.

3.2.9.2 Steel castings - Steel castings shall conform to Specification MIL-S-17509.

3.2.9.3 Steel sheet - Steel sheets shall conform to Specification QQ-S-766, Class 304, 304L, 316, 316L, 321, 347 or 348 and Condition A of the class used.

3.2.9.4 Steel rod - Steel rod shall conform to Specification QQ-S-763, Class 304, 304L, 316, 316L, 321 or 347, and Condition A of the class used.

3.2.9.5 Steel wire cable - The steel wire cable used for chin strap slings, shall conform to Specification QQ-W-423, Form 1, Material FS-302. It shall be 3/32 inch in diameter and be coated with nylon conforming to Specification MIL-N-18352. After coating, the diameter of the cable shall be 1/8 inch.

3.2.9.6 Steel springs - The material used for the steel springs shall conform to the commercial designation 17-7-PH for a precipitation hardenable stainless steel and shall be heat treated to a hardness in the range of 43-48 Rockwell "C".

3.2.9.7 Snap fasteners - Snap fasteners shall be Class 2, Style 3, Construction A of Specification MIL-F-10884.

3.2.10 Suspension pads - The material used for the suspension pads shall be flexible polyurethane foam. The density shall be 3 $\pm$ 0.20 pounds per cubic foot. The hardness shall be equivalent to 25 to 45 durometer as measured by a Shore type 00 hardness tester. It shall have an average cell count of 40 $\pm$ 10 cells per inch. It shall have no interior blow holes over 1/8 inch in diameter and be free from large sun bursts (surface voids).

3.2.11 Pad coating - Material used to coat suspension pads shall be a polyvinyl chloride compound.

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3.2.12 Leather

3.2.12.1 The leather for pad covers shall be vegetable tanned, full grain calfskin, free from defects which may affect appearance or serviceability. The leather shall yield the weight, and thickness specified without splitting the skin after tanning, and shall be firm but pliable and free of bony or hard spots. The leather shall conform to the requirements given below when tested by methods specified in 4.7.17.

3.2.12.2 Finish - The leather shall be finished on the grain side with a bright natural finish and the flesh side shall be clean and free from loose fibers. The use of lead salts or chrome pigments is prohibited.

3.2.12.3 Thickness - The thickness of the leather shall not be less than 1-1/2 ounces nor more than 2-1/2 ounces when measured by a Woburn or similar gage.

3.2.12.4 Tensile strength - The tensile strength of the leather shall not be less than 2000 pounds per square inch.

3.2.12.5 Oil - The leather shall contain not less than 6 percent oil on a moisture free basis.

3.2.12.6 Ash - The total ash content of the leather shall not exceed 2.0 percent on a moisture free basis.

3.2.12.7 Acidity - The acidity (ph) of the leather shall not be less than 3.0 nor more than 5.0.

3.2.12.8 Stitch tearing strength - The stitch tearing strength of the leather shall be not less than 18 pounds.

3.2.13 Chin cup - The material for the chin cup shall be a blend of acrilonitrile-butadien-styrene-rubber. The color shall be comparable to color #36270 of FED STD 595. It shall have the following physical characteristics when tested as specified in 4.7.16:

Izod impact strength (ft. lbs/inches of notch) at 70°F	6.5 minimum
Izod impact strength (ft. lbs/inches of notch) at 20°F	3.5 minimum
Izod impact strength (ft. lbs/inches of notch) at 0°F	2.6 minimum
Izod impact strength (ft. lbs/inches of notch) at -20°F	2.0 minimum
Hardness (Rockwell R)	60 to 80
Tensile Strength (psi)	4000 minimum
Compressive Strength (psi)	5000 minimum

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Modulus of Elasticity (psi):	
Tension	180000
Flexure	200000
Modulus of rupture (psi)	6000
Taper abrasion (Weight loss in grams)	.6
Flammability	Self extinguishing

3.2.14 Webbing - Webbing shall conform to Specification MIL-W-530, Type I. The width shall be 3/4 inch and the color shall be comparable to color #36270 of FED STD 595.

3.2.15 Thread - The thread shall be nylon conforming to Specification MIL-T-7087, Type II, Class II, Twist "Z", Size E. The color shall be comparable with color #36270 of FED STD 595.

3.2.16 Dipping compound - The material used to dip the end of cotton tape shall be a cellulose acetate butyrate compound, in liquid form.

3.2.17 Material in contact with the skin - Materials that may have contact with the wearer's skin shall be non-irritating to normal skin.

3.3 Construction - The construction of all components and assemblies of the helmet covered by this specification shall conform to the requirements of this specification and the design, details and dimensions specified on the applicable drawings. After assembly of body and prior to cleaning and finishing, all burrs, rough spots and sharp edges shall be removed from all components.

3.3.1 Ballistic shell - The ballistic shell shall be fabricated of materials specified on Drawing No. 1884629. It shall be free from unfolded glass wrinkles more than 1/8 inch wide, and there shall be no point where glass cloth folds back upon itself. The shell surface shall be free from blow holes more than 1/16 of an inch in diameter or 1/32 of an inch deep. The resin content of the shell shall be 21 percent plus or minus 3 percent by weight based on total weight of shell. The contour of the shell shall conform to Drawing No. 1884629 and the thickness shall conform to Drawing No. 1884628. The weight of the trimmed, but unpainted bare shell shall not be less than 37 ounces nor exceed 40 ounces.

3.3.2 Metal components - Metal components, in addition to the requirements of the applicable drawings, shall be free from any visible cracks or splits.

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3.4 Assembly - The complete helmet assembly shall consist of a doron ballistic shell, a sound attenuating liner, a fore and aft suspension system, a suspension sizing device, and a chin strap with a safety release device. The helmet shall be assembled in the following manner, although not necessarily in the order stated, to conform with the requirements of the applicable drawings.

3.4.1 Shell - The shell shall be molded, trimmed and drilled. Exposed surface of the shell shall be filled with the material specified, sanded smooth, and prepared for application of the protective coating.

3.4.2 Sound attenuating material - The sound attenuating material shall be cut the specified size and cemented in place in the shell. After cement has been cured, excess sound attenuating material shall be removed and excess cement cleaned from the assembly.

3.4.3 Suspension - The suspension system, suspension sizing device and chin strap shall be assembled in accordance with applicable drawings and riveted into the shell.

3.4.4 Protective coating - The protective coating shall be applied to all exposed surfaces of the shell, sound attenuating material and suspension rivet heads. No protective coating shall be applied to the suspension assembly, suspension sizing device, chin strap, or portion of suspension mounting rivets inside of the shell.

3.4.5 Stitching - All stitching shall be of type 301 of Specification DDD-S-751, 7 to 9 stitches per inch. All seams will be back stitched at least 1/4 inch.

3.5 Performance characteristics

3.5.1 Ballistic resistance

3.5.1.1 Helmet body shells, without attachments or inner linings shall be tested for penetration and impact resistance, in accordance with 4.7.1 and 4.7.2 using the Caliber .15 (5.85 $\pm$ .15 grain) fragment simulator at 0° obliquity at ambient temperature and shall have mean protection (VP50) ballistic limits as specified in Table I. The VP50 ballistic limit shall be calculated using the formula given in Table I.

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3.5.1.2 Passing performance - A test helmet shall be considered to have passed the ballistic test if the mean protection limit velocity (VP50) is equal to or greater than the tabulated value obtained from Table I for the unit weight under consideration.

3.5.1.3 Failing performance - A test helmet shall be considered to have failed the ballistic test if the mean protection limit velocity (VP50) is less than the tabulated value obtained from Table I for the unit weight under consideration.

TABLE I

Ballistic Test Conducted at Normal Obliquity
Caliber .15 (5.85 ± .15 Grain) Fragment Simulator

Weight per Unit Area lbs./sq. ft.	Mean Protection Limit Velocity (VP50) Ft./Sec.
1.1	1056
1.2	1082
1.3	1108
1.4	1134
1.5	1160
1.6	1186
1.7	1212
1.8	1238
1.9	1264

Mean Protection Limit Velocity (VP50):

$$VP50 = \frac{\Sigma V + 0.5\,(Ni-Nc)\,(VIH-VCL)}{Ni + Nc}$$

Where ΣV = the sum of the observed striking velocities in the zone of mixed results, i.e., if the velocities are arranged in order of magnitude, the sum of all velocities from the lowest complete penetration to the highest incomplete penetration; Ni, Nc = the number of incomplete and complete penetrations in the zone of mixed results, respectively; VIH = the highest striking velocity resulting in an incomplete penetration and VCL = the lowest striking velocity resulting in a complete penetration.

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3.5.2 Safety release device - The safety release device consisting of a hook and a ball shall have the following characteristics: The applied load necessary to cause separation of the ball from the hook shall be a minimum of 10 pounds and a maximum of 17 pounds when tested as specified in 4.7.7. A release assembly shall be considered to fail where the release load of any of three consecutive tries exceeds 17 pounds or is less than 10 pounds.

3.5.3 Delousing treatment resistance - The helmet assembly shall not be affected (as determined by visual inspection) when exposed for one (1) hour to methyl bromide vapor when tested as specified in 4.7.15.

3.5.4 Flammability - The ballistic shell when tested as specified in 4.7.3 shall be self extinguishing.

3.5.5 Water immersion - The helmet assembly when tested as specified in 4.7.4 shall not increase in weight more than 5 percent over dry weight. After drying for 24 hours, the helmet assembly shall show no visible damage as a result of the water immersion test.

3.5.6 Heat aging - The helmet assembly when tested as specified in 4.7.2 shall show no visible damage as a result of the heat aging test.

3.5.7 Finish - The protective coating shall adhere firmly to the ballistic shell. After the protective coating is completely cured it shall not be affected by immersion in fresh water for 500 hours, immersion in sea water for 100 hours, or immersion in gasoline for 18 hours when tested as specified in 4.7.6.1. The surface of the protective coating shall not flake off between knife scratches when subjected to the knife test in 4.7.6.2. The heat aging test specified in 3.5.6 shall show no effect on the protective coating.

3.5.8 Weight - The weight of the finished helmet shall not exceed 60 ounces.

3.5.9 Adhesion of foam components - It shall not be possible to remove the foam components from the shell or suspension pad supports without destroying them when tested in accordance with 4.7.8. This requirement is applicable both before and after aging and water immersion tests.

3.5.10 Impact resistance - No permanent visible physical damage shall result when the complete helmet is tested as specified in 4.7.5.

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3.5.11 Tensile strength of chin strap sling - The chin strap sling shall be capable of withstanding a static load of fifty (50) pounds when tested as specified in 4.7.13.

3.6 Workmanship - The finished articles shall be clean, well made and free from any defect which may affect appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection - Unless otherwise required by the procuring activity, the contractor is responsible for the performance of all tests and examinations to assure that helmets delivered to the Government are in full accordance with all applicable requirements. In addition, the contractor is required to maintain records of all tests and examinations performed and to make these records available to the Government representative upon request.

4.2 Classification of tests - The inspection and testing of helmets shall be classified as follows:

(a) Preproduction tests (see 4.5)
(b) Acceptance tests (see 4.6)

4.3 Test Conditions - Unless otherwise specified, all measurements and tests shall be made at average room ambient temperature, pressure and humidity.

4.4 Government inspection - The Government representative will examine the contractor's records and processes and will witness or perform examinations and tests as are deemed necessary to assure that components and assemblies of the helmets are in accordance with all applicable requirements.

4.5 Preproduction Tests - Preproduction tests consisting of those tests specified in Table II as well as inspection for applicable defects specified in 4.7 shall be performed at a facility designated by the procuring activity unless otherwise specified, the preproduction sample shall consist of six (6) complete helmet assemblies described in 3.4 and four (4) bare ballistic shells described in 3.3.1. Failure of any helmet in any requirement may, at the discretion of the procuring activity, be cause for refusal to grant approval. Further production of the item prior to approval by the procuring activity shall be at the contractor's risk.

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4.5.1 Preproduction Sample for Subsequent Contracts - The necessity for a preproduction sample will be determined by the procuring activity when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.6 Acceptance Tests - Acceptance tests, consisting of those tests specified in Table II and 4.7, shall be conducted by or under the supervision of a Government inspector, either at the place of manufacture or at another location acceptable to the procuring activity in accordance with procedure specified in 4.7.

4.6.1 Lot size - As applied to Government inspection of units or product the term "lot" shall mean "Inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.6.2 Sampling for acceptance - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in determining the acceptability of helmets submitted for acceptance will be in accordance with the following:

4.6.2.1 Non-destructive tests - Sampling for non-destructive examination and tests, Table II-A, shall be in accordance with MIL-STD-105, with an AQL of 2.5%.

4.6.2.2 Destructive tests - Sampling for destructive tests, Table II-B, shall be in accordance with MIL-STD-105, Inspection Level L4. The order in which the tests are performed should be arranged so that all tests are performed on each sampled unit. If a sample unit fails to meet any of the requirements, the lot shall be rejected.

4.6.3 Sample Unit - Unless otherwise specified, the sample unit of each component shall be the number of pieces of that component required for one complete helmet.

4.6.4 Action in Case of Lot Rejection - If a lot is rejected, the action taken shall be in accordance with MIL-STD-105.

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TABLE II

ACCEPTANCE TESTS

A - Non-Destructive

Component	Characteristic	Requirements	Test Method
Complete Helmet	Workmanship	3.6	Visual
Complete Helmet	Weight	3.3.1 & 3.5.8	4.7.9
Complete Helmet	Delousing Treatment	3.5.3	4.7.15
Chin Strap Assembly	Safety release device	3.5.2	4.7.7
Chin Strap Assembly	Swage test	3.5.11	4.7.13

B - Destructive

Complete Helmet	Protective coating	3.2.7 & 3.5.7	4.7.6
Complete Helmet	Immersion Test	3.5.7	4.7.6.1
Complete Helmet	Knife Test	3.5.7	4.7.6.2
Complete Helmet	Heat Aging	3.5.6	4.7.2
Complete Helmet	Water Immersion	3.5.5	4.7.4
Complete Helmet	Adhesion of foam components	3.2.5 & 3.5.9	4.7.8
Complete Helmet	Impact resistance	3.5.10	4.7.5
Ballistic shell	Ballistic limit	3.5.1	4.7.1
Ballistic shell	Resin content	3.3.1	4.7.14
Ballistic shell	Flammability	3.5.4	4.7.3
Chin cup	Physical properties	3.2.13	4.7.16 (Table III)
Leather covers	Physical properties	3.2.12	4.7.17 (Table IV)

4.7 Test Methods

4.7.1 Ballistic Test - Helmet samples submitted for qualification shall be subjected to ballistic tests at ambient temperature and after exposure to the heat aging test conducted in accordance with 4.7.2. If the helmet fails the test at ambient temperature, the other test shall not be performed.

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4.7.1.1 Helmet samples for acceptance shall be subjected to ballistic tests at ambient temperature only.

4.7.1.2 The ballistic test shall consist of impacts using the Caliber .15 (5.85 ± .15 grains) fragment simulator at 0° obliquity and mean protection (VP50) ballistic limits as specified in Table I. The VP50 ballistic limit shall be calculated using the formula given in Table I. Prior to commencing the penetration test, the helmet surface shall be marked off into at least 40 blocks of approximately equal size. Twenty-five blocks shall be selected at random and thickness measurements to 0.001" taken at the center of each block. The mean weight per unit area over the selected blocks shall be calculated in pounds per square foot to the nearest 0.01 pound. A ballistic testing fixture with electronic counter chronograph for measuring velocities shall be employed. One round shall be fired at the approximate center of each block with striking velocities varied in order to obtain both complete and incomplete penetrations in a velocity bracket not to exceed 100 feet per second. Mean protection limit velocities (VP50) shall then be calculated. VP50's for weights not listed in Table I shall be obtained by straight line interpolation between the tabulated values. A penetration shall be considered to be "complete" for the purpose of this specification if the fragment simulator passes through the helmet shell or if a fragment from the helmet or the fragment simulator produces a hole in a witness plate of 0.020 2024-T aluminum alloy placed parallel to the inside of the helmet and approximately 2" behind the point of impact. Any round shall be disregarded which, in the judgment of the officer conducting the test, does not represent a valid test. Also, any round shall be disregarded if impact occurs within one inch of a former impact as measured from center to center of impacts, or within one inch of the rim of the helmet shell.

4.7.2 Heat Aging - One of the two helmet shells submitted for qualification shall be subjected to the following conditions in the order listed:

Initial humidification consisting of twenty-four hours at 150 degrees, plus or minus 5 degrees F at a relative humidity of 95 to 100 percent.

Immediately following the initial humidification, the helmet shall be held for twenty-four hours at minus 70 degrees, plus or minus 5 degrees F in air. Then the helmet shell shall be again

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held for twenty-four hours at 150 degrees plus or minus 5 degrees F at a relative humidity of 95 to 100 percent. Immediately following complete cycling the helmet shall be allowed to return to ambient temperature and then tested ballistically in accordance with 4.7.1.

4.7.3 Flammability - Specimens for this test may be cut from any portion of the shell most convenient. The test shall be conducted in accordance with method 2021.1 of Specification L-P-406.

4.7.4 Water Immersion Test - The test shall be conducted on a helmet assembly with the pad covers and suspension pads removed. The sample shall be conditioned for 24 hours at 73.5°F ± 2°F and at a relative humidity of 50% ± 3%. The sample shall be weighed. It shall then be completely immersed in water at a temperature of 73.5°F ± 2°F for a period of 24 hours, removed from the water and any excess water blotted off. The sample shall then be reweighed and the percentage change calculated based on the weight before immersion. Any change in shape shall be noted.

4.7.5 Impact Resistance Test - The helmet shall be placed on a wooden head form in a position similar to that when worn on a man's head, and the chin strap fastened. An eight (8) pound steel or iron ball shall then be dropped vertically from a distance of five (5) feet (equivalent to 40 foot-pounds) to strike the helmet in the center of the crown. The distance shall be measured from the top of the shell to the bottom of the ball.

4.7.6 Protective Coating - The protective coating shall be allowed to air dry for a minimum of one week prior to performance of the tests listed below. The temperature of the various liquids used for the immersion test shall be held to 73°F ± 2°F for the duration of the test. Different samples of the coating shall be used for each of the immersion tests.

4.7.6.1 Immersion Test - The test specimen shall be completely immersed in the various liquids for the number of hours specified in 3.5.7. After the immersion time has elapsed, the sample shall be removed, excess liquid blotted off and the coating examined for any damage.

4.7.6.2 Knife Test - The point of a sharp knife held at approximately 30° to the surface of the test specimen shall be drawn across that surface using that force necessary to scratch through the coating. The procedure shall be repeated several times

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to produce scratches approximately 1/16 to 1/8 inch apart and parallel to each other. The specimen shall then be turned 90° and the knife again used to make another series of scratches intersecting the first series. The specimen shall then be wiped clean of any small paint chips remaining. The coating shall be considered to fail if it has chipped off in the area of the squares resulting from the knife scratches.

4.7.7 Release Test of Safety Release Device - The safety release shall be tested by measuring the static load necessary to cause the ball and hook to separate from the joined and fully seated position. The joined hook and ball shall be pulled from the fully seated position thirty (30) times and then placed in a tension testing fixture. A uniform tension load shall be applied to the joined assembly. Each assembly shall be subjected to three consecutive pulls to effect separation.

4.7.8 Adhesion of Foam Components - All pads shall be tested for adhesion before and after heat aging. The sound absorbent material shall be tested after water immersion and heat aging. The test shall be performed by grasping the foam with the fingers and forcing a knife blade under the foam in order to pry edge loose. After approximately 1/2 inch is pried loose, grasp with thumb and forefinger and attempt to pull remaining foam loose from backing.

4.7.9 Weight - All weight tests shall be performed on a balance scale accurate to 1/4 ounce.

4.7.10 Visual Tests - All visual tests shall be performed in a well lighted area.

4.7.11 Hardness Test - Hardness test shall be performed on a Superficial Rockwell hardness tester or its equivalent.

4.7.12 Measurements - Measurements shall be performed with any suitable measuring device.

4.7.13 Swage Test - The swaged hanger on the chin strap shall be tested by suspending the test sample by one hanger and applying a static load of fifty (50) pounds to the other end. If either swage fitting pulls out, it shall be considered that the sample has failed.

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4.7.14 Resin Content Test - The specimen used for this test shall be a trimmed, bare shell conditioned for 24 hours prior to the test at a temperature of 73°F ± 2°F and a relative humidity of 50% ± 3%. The resin content shall be determined in accordance with Federal Specification L-P-406b, Method 7061.

4.7.15 Delousing Test - The specimen for this test shall be a complete helmet assembly. The specimen shall be placed in a closed 5 cubic foot chamber and exposed to methyl bromide vapor for one hour. The concentration of the vapor shall be that obtained when 40 ml. of liquid methyl bromide (SP. GR. 1.72) is allowed to evaporate in the chamber.

4.7.16 Chin Cup Materials - The chin cup material testing shall be performed as specified in Federal Specification L-P-406, using methods listed in Table III.

TABLE III

Test	Method (Fed. Spec. L-P-406)
Impact Strength	1071
Hardness	1081
Tensile Strength	1011
Compressive Strength	1021
Modulus of elasticity - Tension	1011
Modulus of elasticity - Flexure	1031
Modulus of rupture	1031
Taper abrasion	1091 (H-18 wheel, 2000 cycles, 1000 gram wt.)
Burning rate	2021

4.7.17 Leather Test Methods - The testing of leather shall be carried out as specified in Federal Specification KK-L-311, using methods listed in Table IV.

TABLE IV

Leather, Chemical and Physical	Method (KK-L-311)
Thickness	1011
Tensile Strength	2011
Stitch tearing strength	2151
Grease (chloroform extract)	6311
Total ash	6421
Acidity (ph)	6621

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5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A - Each helmet shall be packaged in an individual box as specified on Drawing 2177070 The box shall be fitted with a corrugated fiberboard insert, cut to fit snugly into the box in the position shown on the drawing. A hole shall be cut in the insert of such size and shape that the helmet is held securely. The box dimensions shown on Drawing 2177070 may be adjusted by the contractor with care being exercised to see that there is a minimum of 1/4 inch clearance between the edge of the ballistic shell and the inside of the box. A fiberboard cell shall be provided for each helmet. The dimension of the cell shall be determined by the contractor. The cell shall be positioned over the ear section of the helmet to act as a spacer and shall provide a snug fit for the helmet in the box.

5.1.2 Level B - Not applicable

5.1.3 Level C - Not applicable

5.2 Packing

5.2.1 Level A - The helmets packaged as specified in 5.1 shall be stacked as specified on Drawing 2177070 in a close fitting heat-sealed vapor barrier bag and then shall be packed in a snug fitting shipping container conforming to the requirements of Specification PPP-B-636, Type 1 or 2, Class 2, Grade 3, PPP-B-601, overseas type, Style A or PPP-B-621, Class 2, Style 2 or 4, Type 2. The gross weight of the wood boxes shall not exceed 200 pounds. The gross weight of the fiberboard containers shall not exceed 70 pounds.

5.2.2 Level B - Not applicable

5.2.3 Level C - Not applicable

5.3 Marking - The interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The helmet described herein is used primarily for ballistic protection for the head of telephone talker personnel. It may be used for ballistic protection or a head cover for other personnel if desired.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction samples required (see 3.1).
(c) Levels of Packaging and packing (see 5.1).
(d) Application of the heat-sealed vapor barrier shall be excluded from helmets when it is known that they are for direct shipment to the fleet for issue.

6.3 Materials - The following materials have been found to meet the requirements of materials specified in paragraphs indicated:

3.2.3 Selectron 5003 Polyester Resin, manufactured by Pittsburgh Plate Glass Co., Pittsburgh, Penna.

3.2.4 Martin Zenour, #6385 High Speed Glazing Putty manufactured by Sherwin-Williams Co., 2520 S. Quarry Road, Chicago, Ill.

3.2.6 Resiweld #4, manufactured by H. B. Fuller Co., Linden, N. J. 3M, EC 1300, manufactured by Minnesota Mining and Manufacturing Co., St. Paul 6, Minnesota.

3.2.7 Rexclad "R", manufactured by Rexton Finishers, Irvington, New Jersey.

3.2.11 Liquid Envelope, White, Vinyl #34-62, manufactured by Better Finishes and Coatings, Inc., Newark, New Jersey.

3.2.13 Royalite, manufactured by U. S. Rubber Co., Chicago 39, Ill.

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